

DTSC 5900

Special Topics in Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (13.0 / 100)	10.8 / 100

11

CURRENT AVG SCORE
across 6 job roles

59

POTENTIAL SCORE
with recommended additions

+60

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Examines special topics in Data Science.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	13.0	39.0	+26.1	MEDIUM
AI Engineer	12.5	55.6	+43.1	HIGH
Machine Learning Engineer	12.0	61.5	+49.5	HIGH
MLOps Engineer	10.9	62.9	+52.0	HIGH
ML Engineer	10.0	69.4	+59.4	HIGH
Applied Scientist	6.7	66.2	+59.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS



This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Data Scientist 13.0 → 39.0 (+26.1 pts)

presentation different

AI Engineer 12.5 → 55.6 (+43.1 pts)

models learning design new tools make

Machine Learning Engineer 12.0 → 61.5 (+49.5 pts)

learning tools design development models applications analysis statistical

MLOps Engineer 10.9 → 62.9 (+52.0 pts)

scientists learning media tools models development

ML Engineer 10.0 → 69.4 (+59.4 pts)

media learning models world new tools

Applied Scientist 6.7 → 66.2 (+59.5 pts)

computational research models department learning industry design development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.